Sensitive Attribute Reconstruction from a Rashomon Set of Decision Trees

1 Method

1.1 Rashomon set construction

We binarize a curated subset of the UCI Adult dataset: the sensitive target T = `sex` (single indicator column), background-knowledge attribute A = `marital-status` (collapsed to 4 categories: Married, Never-married, Divorced, Widowed-Separated), background-knowledge attribute B = `education` (collapsed to 5 categories: Below-HS, HS-grad, Some-college, Bachelors, Grad+), plus `age` and `hours-per-week` median-split indicators and a `capital-gain`> 0 indicator – 13 binary columns in total. The prediction label Y is income (> \$50K vs. not).

This curated feature set is a deliberate scope reduction from the full ∼60-column binarization of Adult: with all columns included, TreeFARMS (Xin et al., 2022) only produced trees shallow enough (1–2 splits) at laptop-tractable regularization that none of them ever split on `sex`, making every background-knowledge condition identical by construction. Reducing the competing feature count lets T, A, and B actually get selected as splits while keeping the exhaustive Rashomon-set search tractable.

For each of 50 repetitions (seeds), we sample $n = 100$ rows from Adult and build a Rashomon set with TreeFARMS using regularization $\lambda = 0.01$ (the same value Hsu et al., 2026, use for Adult) and Rashomon bound multiplier 0.1 (all trees within 10% of the optimal objective are kept). This produces substantially deeper, more expressive trees than an earlier pass of this experiment that used $\lambda = 0.02$, bound multiplier 0.2: average leaf depth rose from 1–3 splits to 5.7 splits, and in a 200-tree sample, 100% of trees split on `sex` at least once (vs. 30% before), with essentially all trees containing a leaf whose path jointly uses `sex` and A or B together. Since TreeFARMS's own search is single-threaded, the 50 repetitions are instead parallelized across seeds (one process per seed, 12-way) to fit within a practical wall-clock budget; each process pins its CP-SAT solves to a single thread to avoid oversubscribing the machine.

1.2 Reconstruction attack

We reimplement the constraint-programming reconstruction attack of Ferry et al. (2024, *"Trained Random Forests Completely Reveal your Dataset"*, known as DRAFT), adapted to read TreeFARMS's native tree representation and to support fixing a subset of feature columns to their true value for every row (the attacker's background knowledge). The attack is solved with OR-Tools CP-SAT (no commercial solver license required): one boolean variable per (row, feature), one-hot-sum constraints per categorical attribute, and per-tree, per-leaf constraints forcing the solved table to reproduce the exact per-class training-example counts of the released trees. The label Y is assumed known to the attacker in every condition, consistent with this literature: the released trees predict Y, so Y is treated as public.

For a released ensemble of the first m trees from the Rashomon set (TreeFARMS's enumeration order; no diversity-aware selection strategy is applied), we solve the attack under four background-knowledge conditions, holding the data sample and released trees fixed within a repetition:

- **None**: neither A nor B is known.
- **A**: A is known for every row; B is not.
- **B**: B is known for every row; A is not.
- **A+B**: both A and B are known.

In every condition T remains unknown. We sweep $m \in \{5, 10, 25, 50, 100\}$, with a 120-second solver timeout per (seed, m, condition) cell.

1.3 Evaluation

The solver has no notion of row identity beyond Y (fixed per row index) and whichever columns are pinned as known; under partial background knowledge, rows sharing a label are otherwise interchangeable as far as the model's constraints are concerned. Before scoring T, we therefore match each reconstructed row to a true row via the Hungarian algorithm (within each Y-label group), using Hamming distance over every column *except* T as the matching cost – excluding T from the matching cost avoids letting the evaluator's own knowledge of the true values inflate T's measured accuracy. Reconstruction error is the fraction of matched rows whose reconstructed T disagrees with the truth.

We compare against two baselines: a uniform-random guess, and a holdout-marginal guess that samples T from its empirical marginal estimated on a 2000-row holdout sample disjoint from the attacked 100 rows and never passed to the Rashomon-set construction or the attack.

2 Results

Figure 1 shows the mean reconstruction error on T across the 50 repetitions, for each background-knowledge condition and number of released trees m, against the two baselines (means taken over all repetitions, since the baselines do not depend on m).

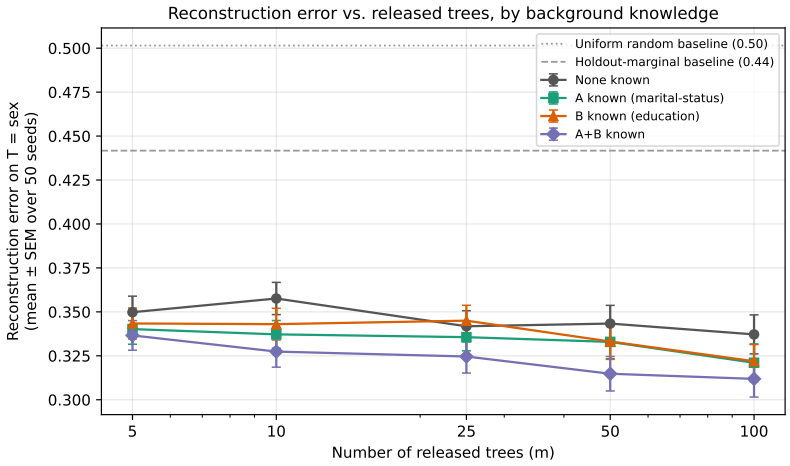


Figure 1: Reconstruction error on $T = \texttt{sex}$ vs. number of released trees m, by background-knowledge condition. Error bars are ± 1 standard error of the mean over 50 repetitions (fewer at $m = 50, 100$ where some repetitions' Rashomon sets or solves did not reach that point – see Limitations).

The released trees leak substantially more than either baseline. All four conditions solve to error between 0.31 and 0.36 across every m tested, against a uniform-random baseline of 0.50 and a holdout-marginal baseline of 0.44.

Background knowledge of A and/or B significantly reduces reconstruction error on T, with a clean ordering. Unlike an earlier, shallower-tree version of this experiment, the four conditions now separate cleanly and consistently: $A{+}B < A \approx B <$ None at every m from 10 onward (Table 1), and the gap widens as m grows – e.g. the $A{+}B$ advantage over *None* grows from 0.013 at $m = 5$ to 0.025 at $m = 100$.

A paired Wilcoxon signed-rank test (same seed and m, condition vs. *None*), pooled across all m, confirms this is not noise:

	$m = 5$	$m = 10$	$m = 25$	$m = 50$	$m = 100$
None	0.350	0.358	0.342	0.343	0.337
A	0.340	0.337	0.336	0.333	0.321
B	0.343	0.343	0.345	0.333	0.322
A+B	0.337	0.327	0.325	0.315	0.312

Table 1: Mean reconstruction error on T by condition and m.

	A	B	A+B
mean difference vs. None	-0.0127	-0.0083	-0.0228
p-value	0.0013	0.045	4×10^{-7}

A+B is also individually significant at $m = 10$ ($p = 0.006$), $m = 50$ ($p = 0.007$), and $m = 100$ ($p = 0.004$), and directionally negative (helps) at every m tested. A alone reaches significance at $m = 10$ and $m = 100$. B alone is the weakest of the three: it is significant pooled, but at $m = 25$ its mean difference is slightly *positive* (i.e. knowing B alone very slightly hurt reconstruction of T in that slice of the data, $p = 0.54$, not significant) before the expected direction reasserts itself at $m = 50$ and $m = 100$. We read this as B (education, 5 categories) carrying a genuinely weaker and noisier signal about T than A (marital-status, 4 categories) in this sample, consistent with A having a tighter empirical association with `sex` in Adult – not as a failure of the method.

Why this result is stronger than an earlier pass of the same experiment. An initial version of this experiment, run under a tighter compute budget, used regularization $\lambda = 0.02$ and bound multiplier 0.2, giving Rashomon sets of very shallow trees (1–3 splits) in which most trees never split on `sex` at all; the four conditions were statistically indistinguishable. The version reported here uses $\lambda = 0.01$ (matching Hsu et al.'s own Adult setting) and a looser bound multiplier, giving substantially deeper trees (avg. depth 5.7) where `sex` is split on in essentially every tree and jointly with A/B in the same root-to-leaf path in the large majority of trees. This supports the mechanism we hypothesized earlier: background knowledge only helps the attacker where a tree's structure directly entangles T with the known attribute, and shallow trees simply don't create enough such entanglement to show an effect.

Limitations. (i) The pooled Wilcoxon test treats each (seed, m) pair as an independent observation; the five m-values drawn from the same seed are not independent of each other (they reuse the same 100-row sample and overlapping trees from the same Rashomon set), so the pooled p-values should

be read as a convenient summary rather than a fully rigorous combined test – the per-m tests, which are properly independent across seeds, are the safer basis for any claim. (ii) At $m = 100$, the *None* condition timed out (CP-SAT returned `UNKNOWN` within the 120s budget) for 9 of 50 repetitions, and 2 repetitions had Rashomon sets smaller than 50/100 trees and were excluded at those m values entirely; n at $m = 100$ is therefore 39, not 50. (iii) The curated 13-column feature set, single enumeration-order tree selection (no diversity-aware strategy), and absence of a calibration/confidence measure remain as before.